

QUESTION:

Write a C program to display the following patterns using nested loops without using arrays:

1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated Number Pyramid

Display a menu for the above patterns and use a switch statement to execute the pattern selected by the user. Implement each pattern using a separate parameterless function with no return value.

Aim

To write a C program using a switch statement and nested loops to display different pyramid and diamond patterns based on the user's choice.

Algorithm

1. Start.
2. Display the menu containing the five pattern choices.
3. Input the user's choice.
4. Use a switch statement to select the required pattern.
5. Input the number of rows n.
6. Use nested loops to print the selected pattern.
7. Display the pattern.
8. Stop.

Test Case Table

Test Case	Choice	Rows	Expected Pattern
1	1	5	Full Pyramid
2	2	5	Inverted Pyramid
3	3	5	Diamond Pattern
4	4	5	Palindromic Number Pyramid
5	5	5	Repeated-Number Pyramid

>C program:

```
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--day27.c - Mousepad

60     for(j = 1; j <= n-1; j++)
61         printf(" ");
62
63     for(j = 1; j <= 2*i-1; j++)
64         printf("+");
65
66     printf("\n");
67 }
68
69
70 void invertedPyramid()
71 {
72     int i, j, n;
73
74     printf("Enter number of rows: ");
75     scanf("%d", &n);
76
77     for(i = n; i >= 1; i--)
78     {
79         for(j = 1; j <= n-i; j++)
80             printf(" ");
81
82         for(j = 1; j <= 2*i-1; j++)
83             printf("+");
84
85         printf("\n");
86     }
87 }
88
89 void diamondPattern()
90 {
91     int i, j, n;
92
93     printf("Enter number of rows: ");
94     scanf("%d", &n);
95
96     for(i = 1; i <= n; i++)
97     {
98         for(j = 1; j <= n-i; j++)
99             printf(" ");
100
101         for(j = 1; j <= 2*i-1; j++)
102             printf("+");
103
104         printf("\n");
105     }
106
107     for(i = n-1; i >= 1; i--)
108     {
109         for(j = 1; j <= n-i; j++)
110             printf(" ");
111
112         for(j = 1; j <= 2*i-1; j++)
113             printf("+");
114
115         printf("\n");
116     }
117 }
118
119
120
121
122
123
124
125
126
127
128
129
130
131
132
133
134
135
136
137
138
139 }
140
141 void repeatedNumberPyramid()
142 {
143     int i, j, n;
144
145     printf("Enter number of rows: ");
146     scanf("%d", &n);
147
148     for(i = 1; i <= n; i++)
149     {
150         for(j = 1; j <= n-i; j++)
151             printf(" ");
152
153         for(j = 1; j <= i; j++)
154             printf("%d ", j);
155
156         printf("\n");
157     }
158 }
159
```

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111
112     for(j = 1; j <= 2*i-1; j++)
113         printf(" ");
114     printf("\n");
115 }
116 }
117 }
118
119 void palindromicPyramid()
120 {
121     int i, j, n;
122
123     printf("Enter number of rows: ");
124     scanf("%d", &n);
125
126     for(i = 1; i <= n; i++)
127     {
128         for(j = 1; j <= n-i; j++)
129             printf(" ");
130
131         for(j = 1; j >= 1; j--)
132             printf("%d", j);
133
134         for(j = 2; j <= i; j++)
135             printf("%d", j);
136
137         printf("\n");
138     }
139 }
140
141 void repeatedNumberPyramid()
142 {
143     int i, j, n;
144
145     printf("Enter number of rows: ");
146     scanf("%d", &n);
147
148     for(i = 1; i <= n; i++)
149     {
150         for(j = 1; j <= n-i; j++)
151             printf(" ");
152
153         for(j = 1; j <= i; j++)
154             printf("%d ", j);
155
156         printf("\n");
157     }
158 }
159
```

>TEST CASE -1,2,3 (Compilation and execution):

```
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali:~]$ gcc day27.c -o day27
[priyadarshna@kali:~]$ ./day27
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated Number Pyramid
Enter your choice: 1
Enter number of rows: 5
*****
*
*****

[priyadarshna@kali:~]$ ./day27
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated Number Pyramid
Enter your choice: 2
Enter number of rows: 5
*****
*****
*
*****

[priyadarshna@kali:~]$ ./day27
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated Number Pyramid
Enter your choice: 3
Enter number of rows: 5
*****
*****
*****
*****
*
*****

[priyadarshna@kali:~]$
```

>TEST CASE-4,5 (Compilation and execution):

```
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
~ (priyadarshna@kali) [-]
$ gcc day27.c -o day27
~ (priyadarshna@kali) [-]
$ ./day27
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated Number Pyramid
Enter your choice: 4
Enter number of rows: 5
  1
 2 2
3 3 3
4 4 4 4
5 5 5 5 5
~ (priyadarshna@kali) [-]
$ ./day27
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated Number Pyramid
Enter your choice: 5
Enter number of rows: 5
  1
 2 2
3 3 3
4 4 4 4
5 5 5 5 5
~ (priyadarshna@kali) [-]
$
```